Moringa School

AI Essentials and Workflow Automation Course

Automated Forex Intelligence & Pricing Platform

Week 10 capstone project

Project Report

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1. Project Overview

The Production-Ready Financial Insights Automation System is an AI-powered workflow automation solution developed using n8n to automate the retrieval, processing, validation, summarization, and reporting of financial intelligence data from multiple financial institutions and forex information sources.

The workflow was designed to simulate a production-grade financial intelligence reporting environment where financial data is continuously collected, validated, summarized using AI, and distributed through automated reporting channels.

Unlike a prototype workflow, the solution emphasizes:

- Structured AI outputs
- Operational reliability
- Fault tolerance
- Workflow validation
- Error recovery
- Governance visibility
- Automated reporting and archival

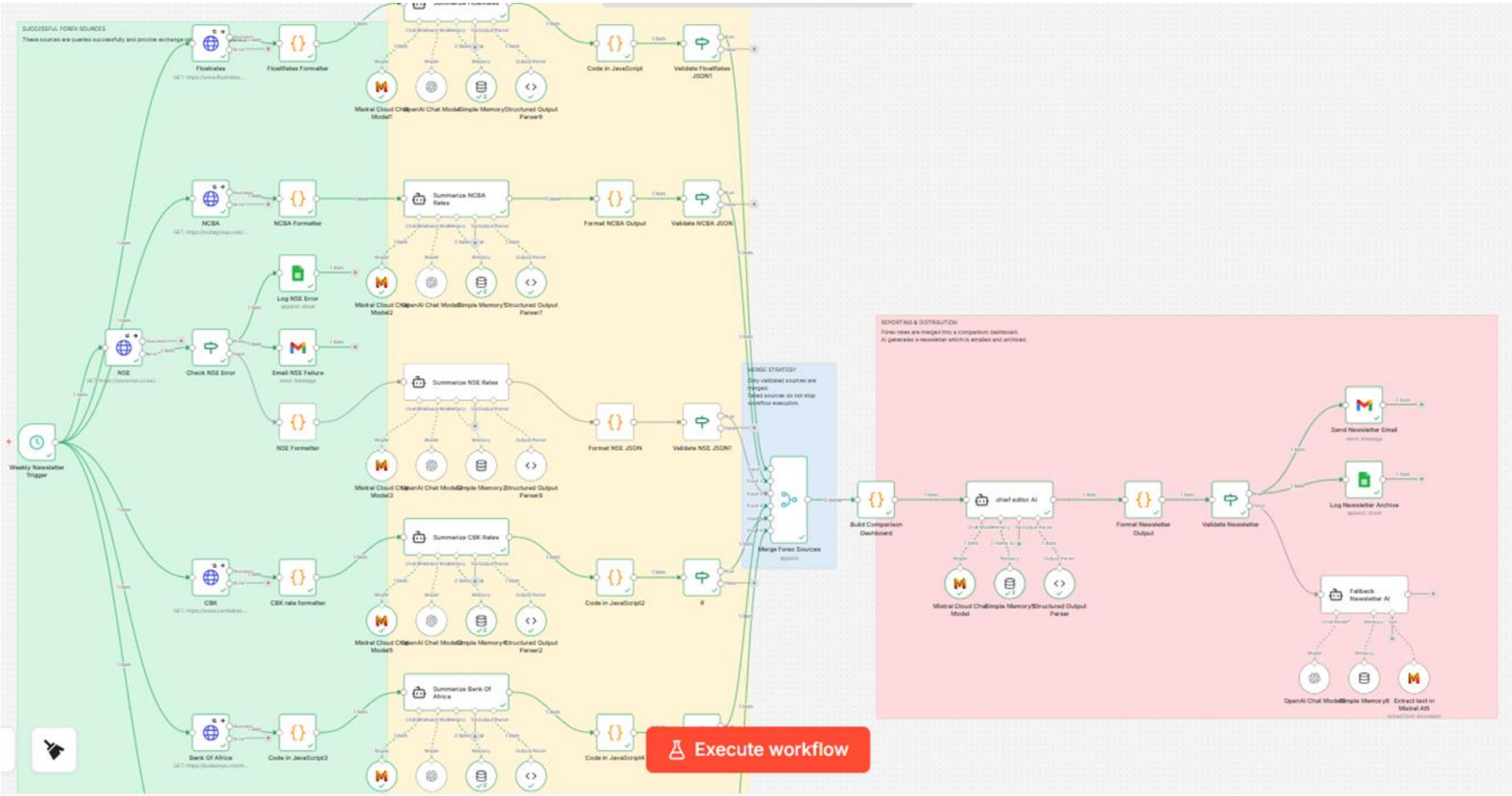
The workflow consolidates financial intelligence from multiple financial sources into a single comparison dashboard and automated financial newsletter.

2. Project Objectives

The key objectives of the project were:

- Provides hourly visibility of market forex rates.
- Enables Treasury to determine competitive customer rates.
- Supports profitability decisions through market benchmarking.
- Reduces manual rate collection activities.
- Improves accuracy through automated validation and error handling.
- Creates a reliable audit trail of historical forex movements.

3. Architecture Diagram



Workflow Zones

The workflow is organized into the following operational zones:

A. Successful Forex Sources Layer

Responsible for retrieving financial intelligence from multiple external sources.

B. Data Extraction & Validation Layer

Responsible for formatting extracted data, AI summarization, structured output parsing, and validation before merge.

C. Merge Strategy Layer

Responsible for consolidating only validated financial intelligence outputs.

D. Reporting & Distribution Layer

Responsible for dashboard generation, newsletter generation, validation, email distribution, and archival.

4. Workflow Explanations & Node Configuration Screenshots

Main Financial Insights Workflow

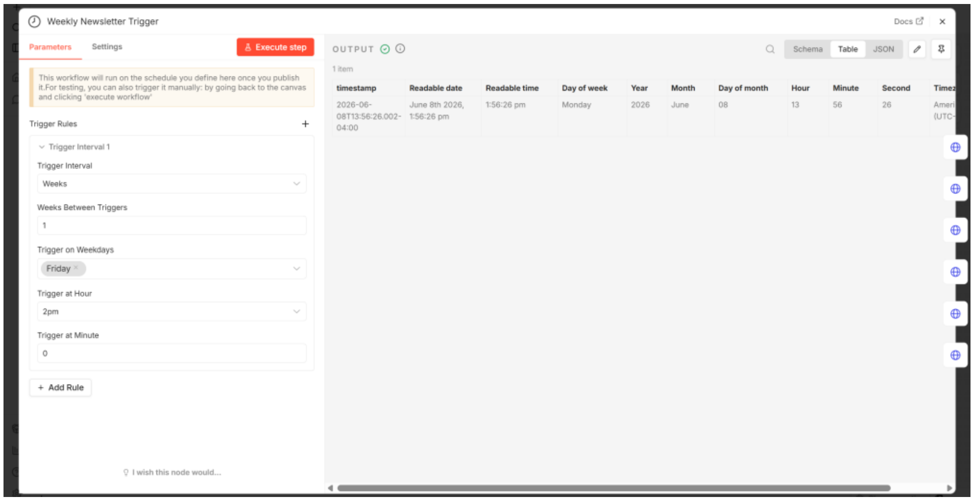
The main workflow performs the following operational steps:

Step 1 – Scheduled Trigger Execution

The workflow begins execution using a scheduled trigger node configured to run automatically.

Purpose:

- Automate periodic execution
- Remove manual intervention
- Simulate production scheduling

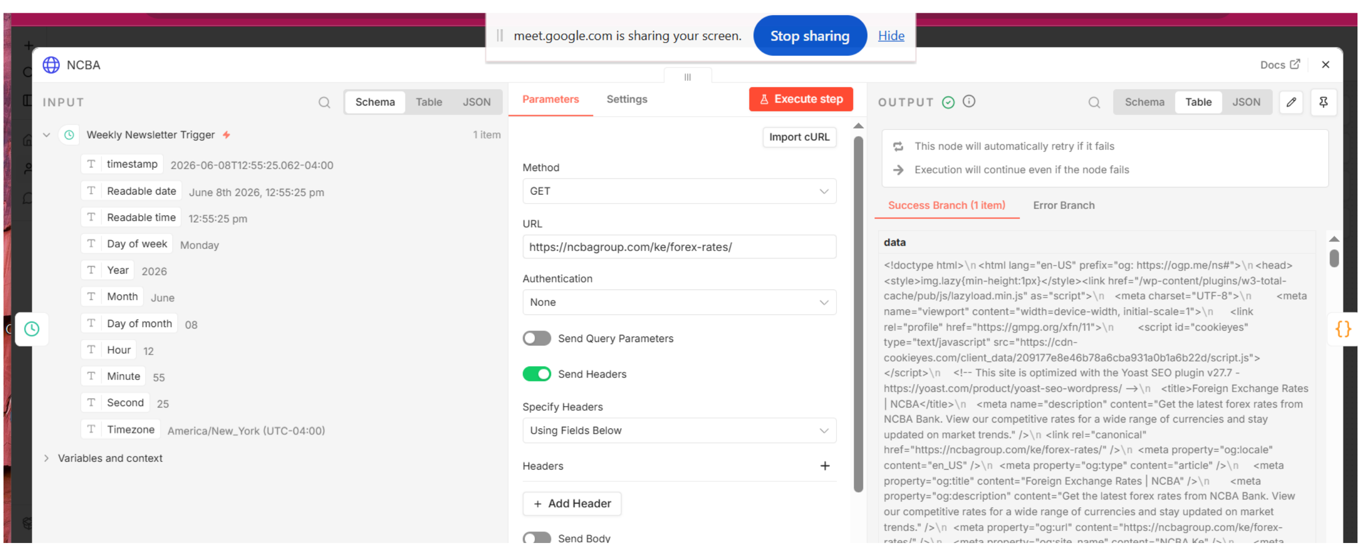


Step 2 – Financial Source Collection (HTTP Request)

The workflow retrieves financial intelligence data from multiple external financial sources including:

- FloatRates
- NCBA
- Nairobi Securities Exchange (NSE)
- Central Bank of Kenya (CBK)
- Bank of Africa
- ExchangeRate API

Each source operates independently to improve fault tolerance and branch-level isolation.



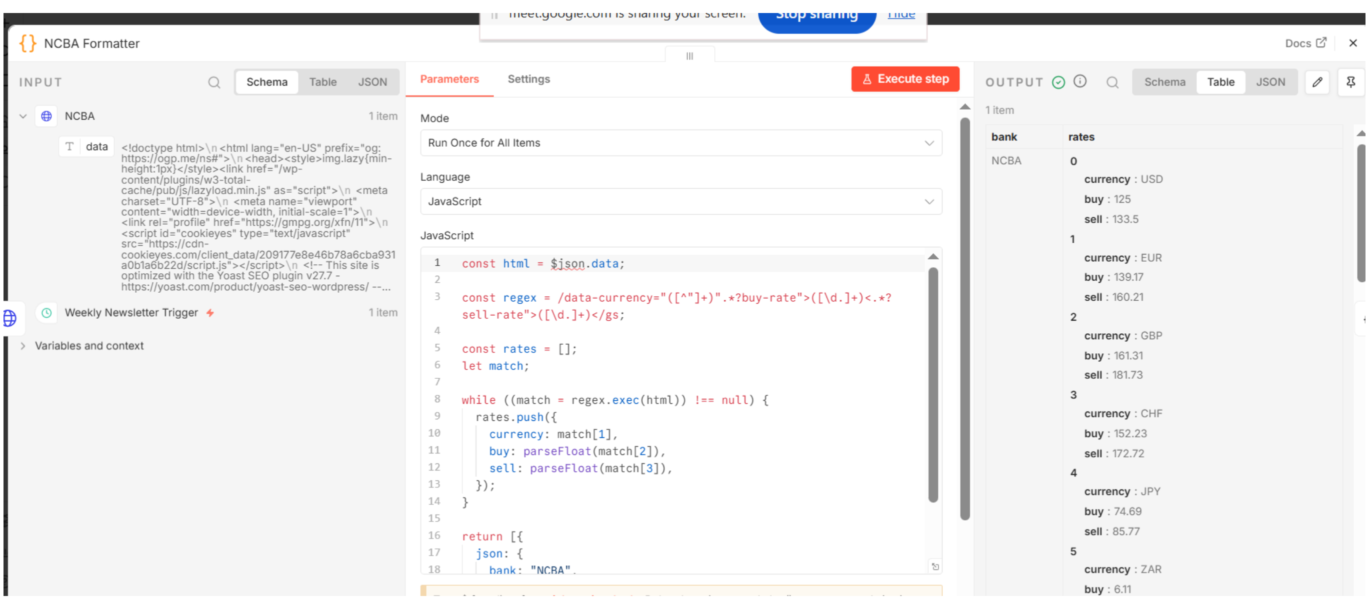
Step 3 – Formatter & Extraction Nodes

Each source passes through a dedicated formatter node responsible for:

- Cleaning raw HTML responses
- Extracting structured financial information
- Preparing content for AI summarization
- Normalizing source outputs

Formatter nodes include:

- FloatRates Formatter
- NCBA Formatter
- NSE Formatter
- CBK Rate Formatter
- Bank of Africa Formatter
- ExchangeRate Formatter



Step 4 – AI Summarization Layer

Each formatted source is processed through a dedicated Mistral AI summarization node.

Purpose:

- Generate financial summaries
- Produce structured outputs
- Normalize source intelligence
- Generate machine-readable outputs

The AI summarization layer uses:

- Mistral AI
- Structured Output Parser nodes
- Memory nodes where applicable

AI Agent User Message

Extract the top forex exchange rates against the Kenyan Shilling (KES).

Return ONLY valid JSON.

source = "FloatRates"

title = "FloatRates Forex Rates"

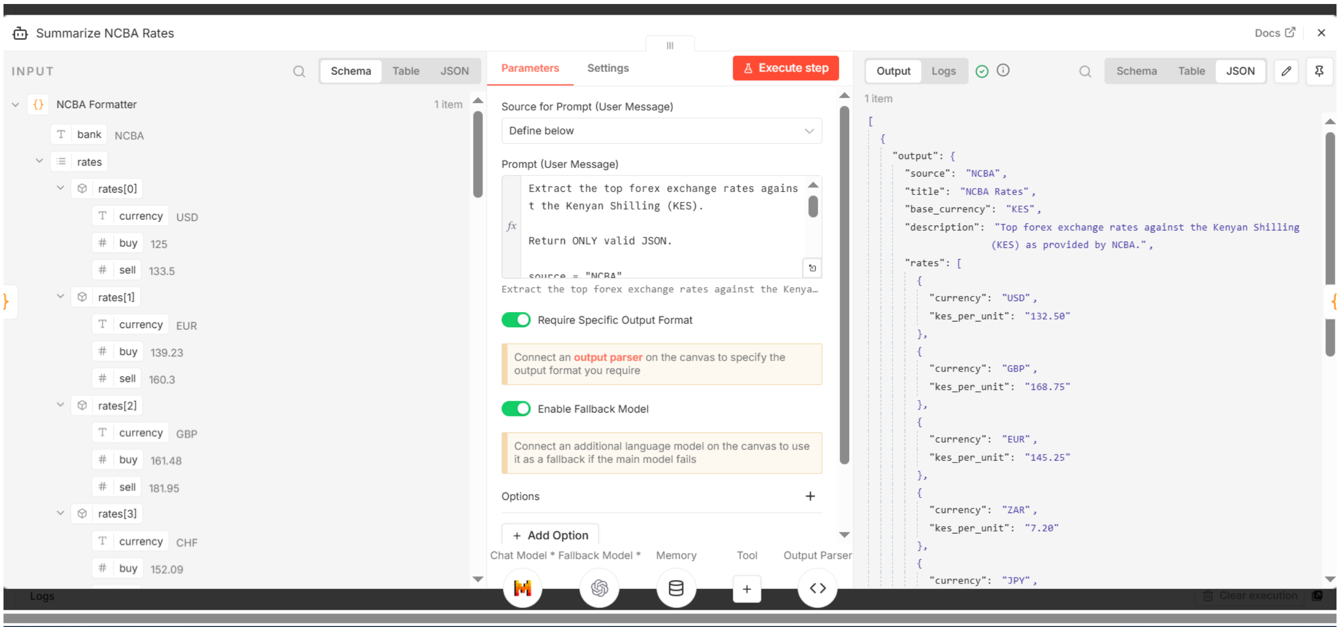
base_currency = "KES"

Include major currencies such as USD, GBP, EUR, ZAR, JPY, AUD, CAD and any other available currencies.

For each currency return:

- currency

- kes_per_unit



Step 5 – Structured Output Validation

After AI summarization, outputs pass through validation nodes.

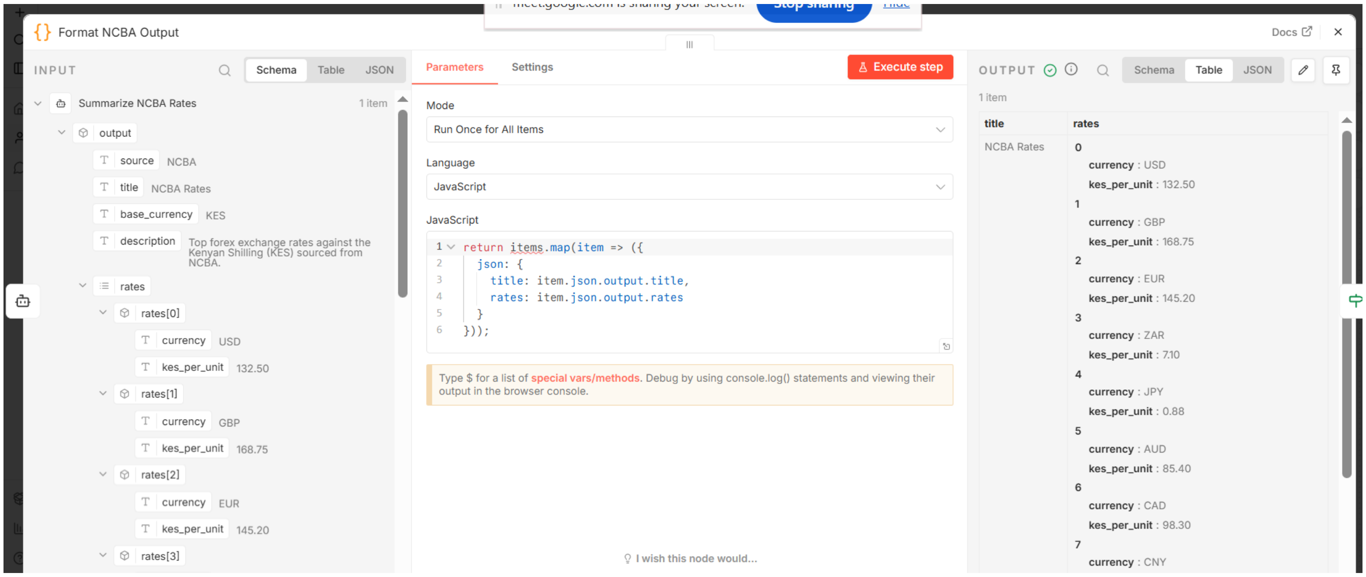
Purpose:

- Prevent malformed outputs
- Enforce structured JSON reliability
- Ensure downstream compatibility
- Improve operational predictability

Validation nodes include:

- Validate FloatRates JSON
- Validate NCBA JSON
- Validate NSE JSON
- Validate ExchangeRate JSON

Validation logic ensures only valid outputs proceed into the merge strategy layer.



Step 6 – Merge Strategy

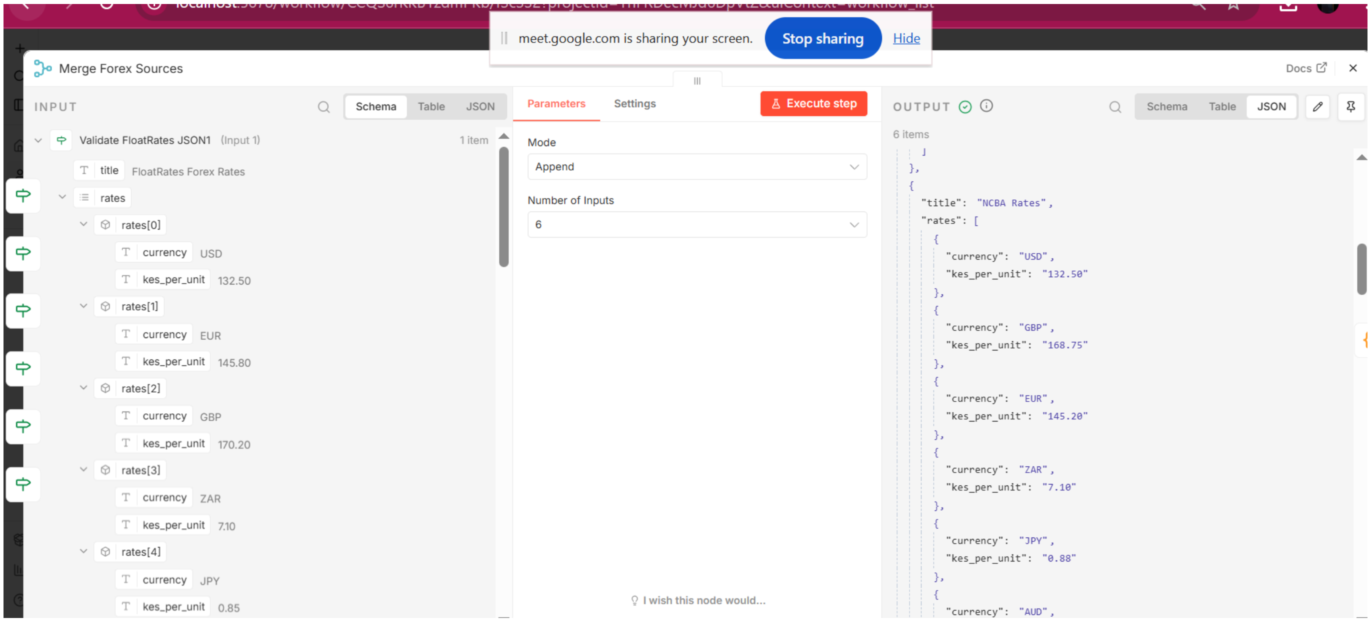
Validated outputs are merged into a centralized workflow path.

Purpose:

- Consolidate validated financial intelligence
- Prevent failed branches from stopping workflow execution
- Implement graceful degradation

The merge strategy ensures:

- Failed sources are isolated
- Workflow continuity is preserved
- Partial success execution is supported

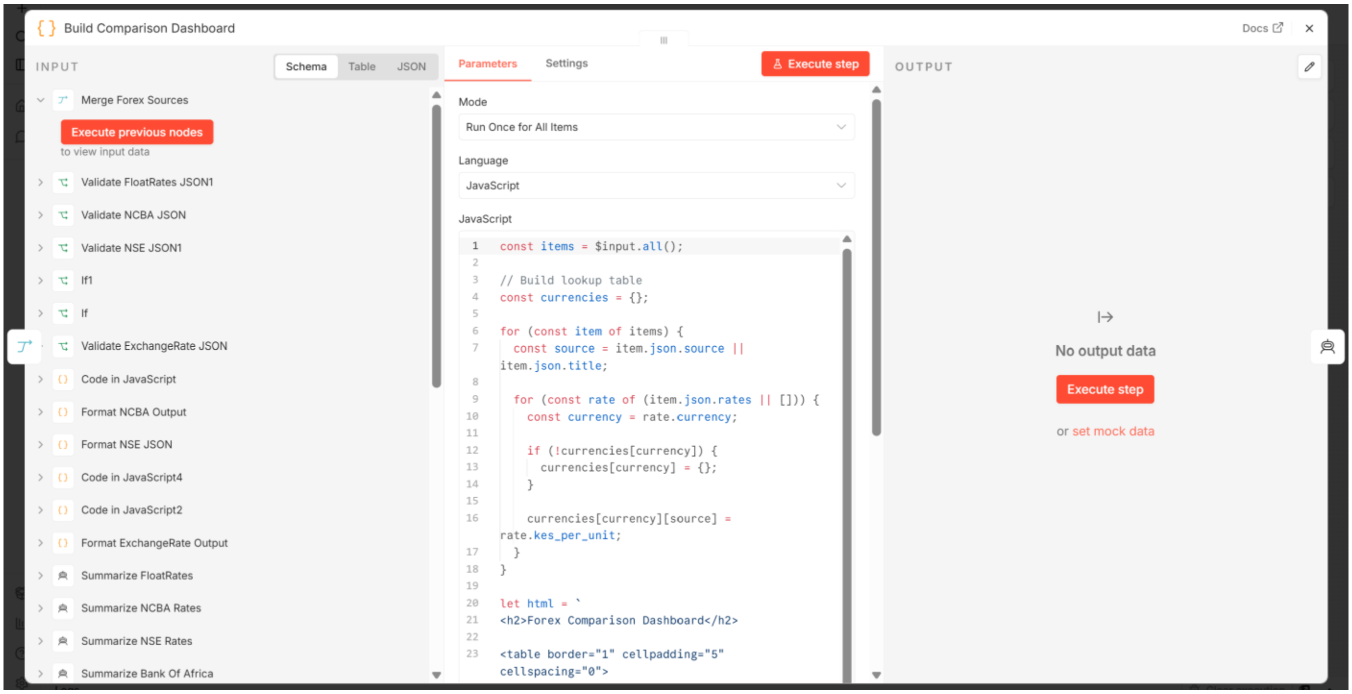


Step 7 – Financial Comparison Dashboard Builder

The workflow consolidates validated forex and financial intelligence data into a unified comparison dashboard structure.

Purpose:

- Create unified reporting structure
- Prepare dashboard table data
- Standardize final reporting outputs



Step 8 – Chief Editor AI

A centralized AI node generates the final financial newsletter using merged financial intelligence data.

Purpose:

- Generate final financial summary
- Produce market outlook
- Consolidate financial insights
- Create final reporting narrative

<u>User Message</u>

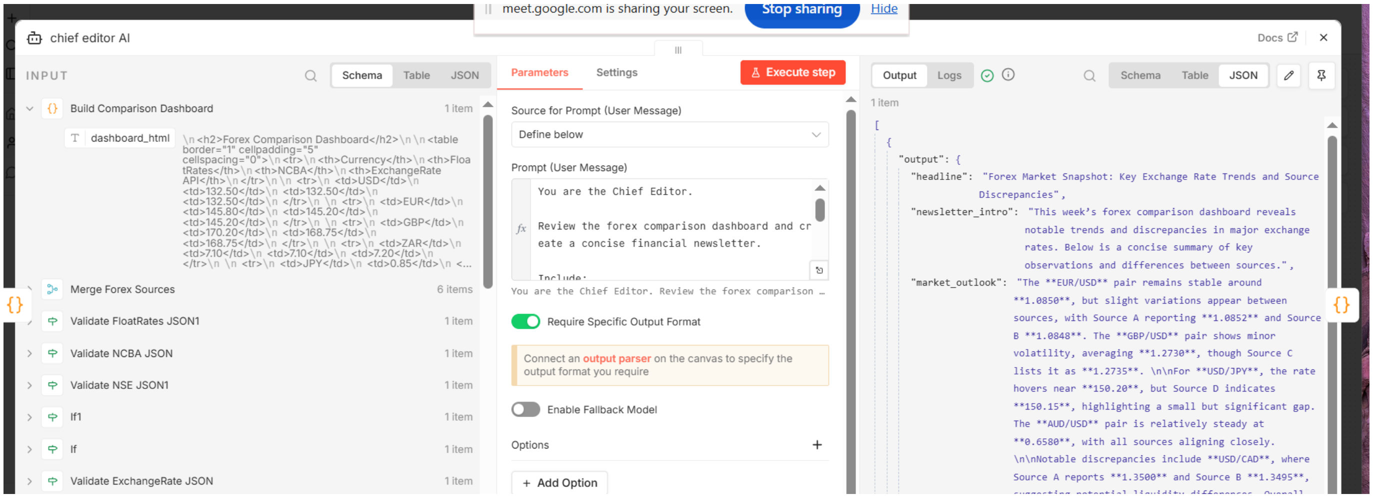
You are the Chief Editor.

Review the forex comparison dashboard and create a concise financial newsletter.

Include:

- *Headline*
- *Brief summary of key exchange rate observations*
- *Mention any notable differences between sources*

Keep the newsletter under 200 words. Return valid JSON only.

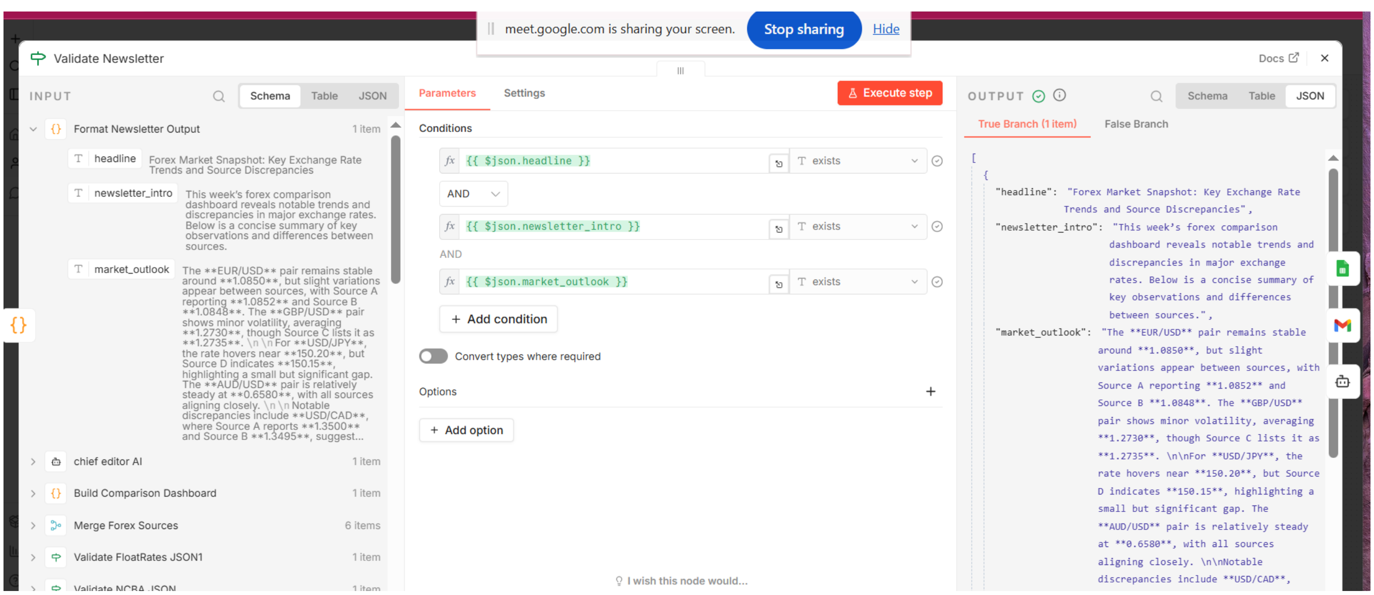


Step 9 – Newsletter Validation

The final newsletter output passes through validation checks before distribution.

Purpose:

- Prevent malformed newsletters
- Improve reliability
- Validate final structured outputs



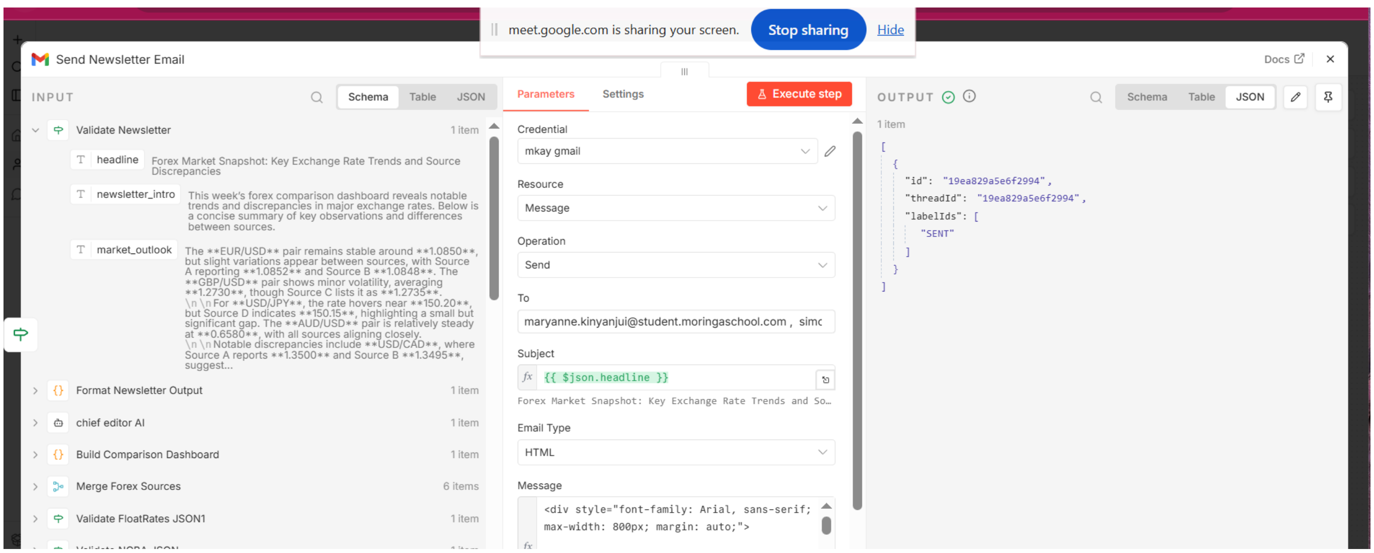
Step 10 – Email Distribution & Archiving

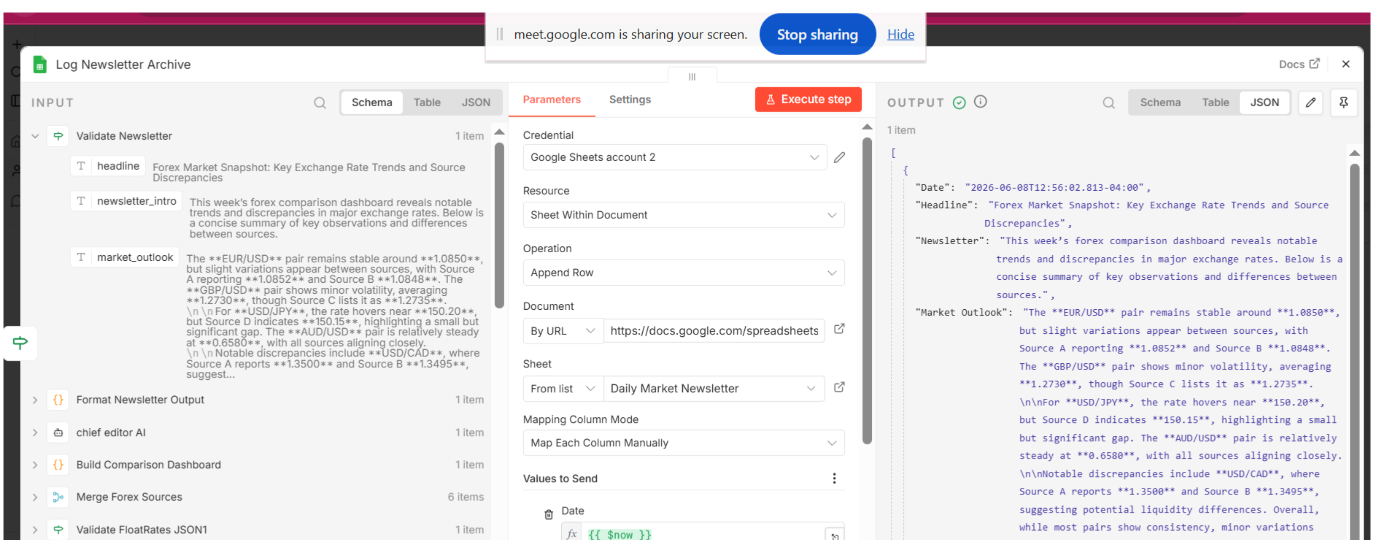
Validated newsletters are:

- Sent through Gmail
- Archived into Google Sheets

Purpose:

- Automated stakeholder distribution
- Governance visibility
- Historical archival
- Audit support





Step 11 – Fallback AI Handling

Fallback AI logic exists to support operational continuity in case of newsletter processing issues.

Purpose:

- Improve resilience
- Support recovery handling
- Prevent complete workflow failure

User Message

Generate a simplified newsletter intro using available summaries.

STRICT RULES:

1. Under 200 words

2. Plain text only

3. Use simple professional tone

4. Return ONLY valid JSON

Input summaries:

{{ *JSON.stringify($json, null, 2)* }}

Return ONLY valid JSON in this format:

{

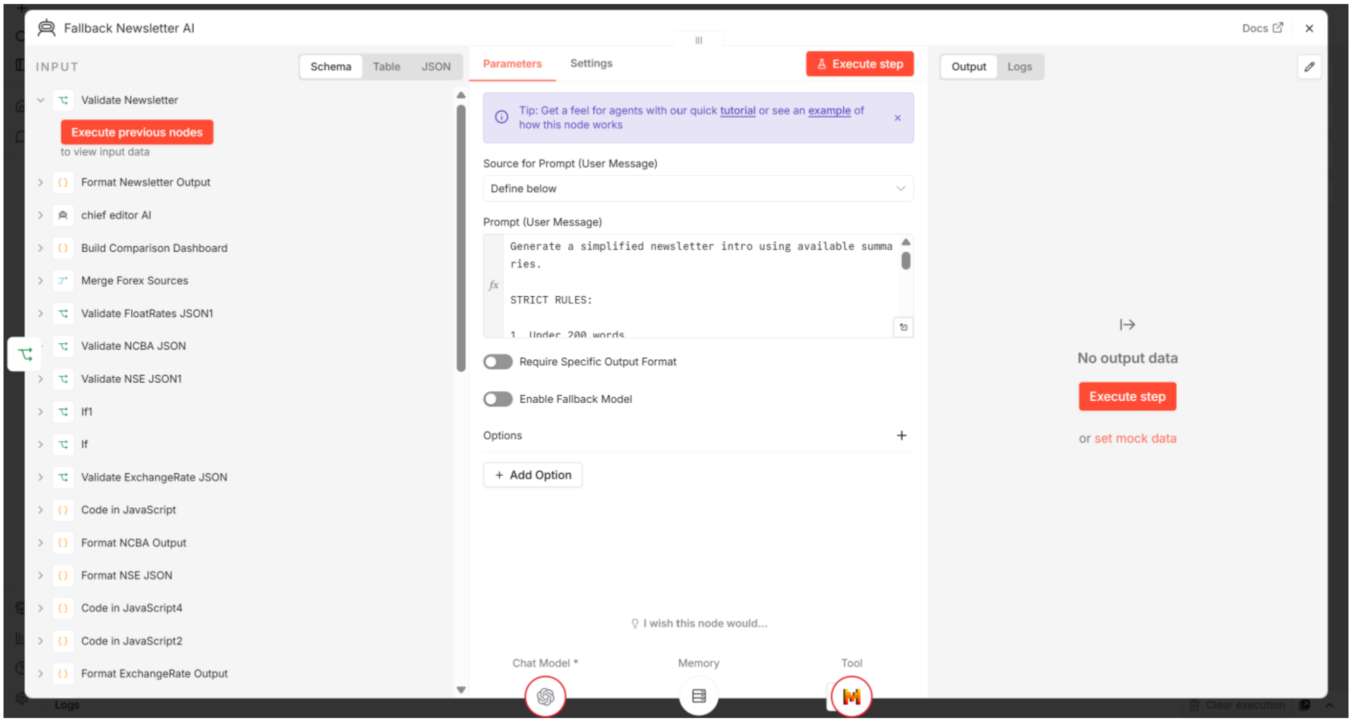
"newsletter_intro": "string",

"word_count": number,

"model_used": "GPT-4o-mini",

"status": "fallback-success"

}



5. Error Handling Workflow

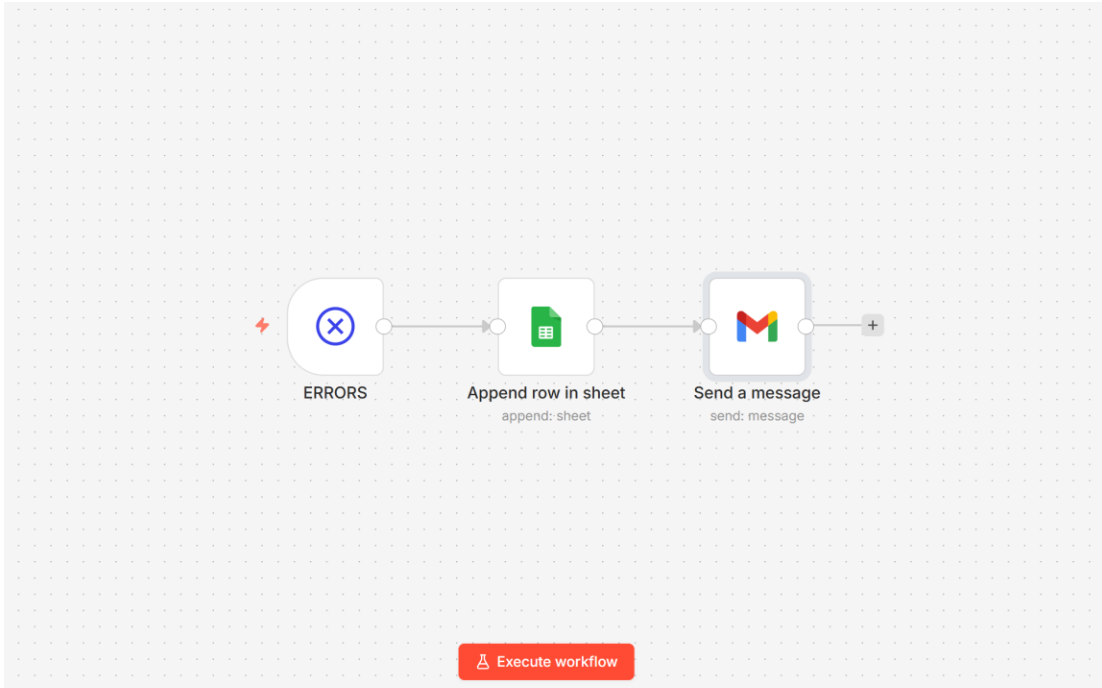
A dedicated Error Handling Workflow was implemented separately from the main workflow.

Purpose:

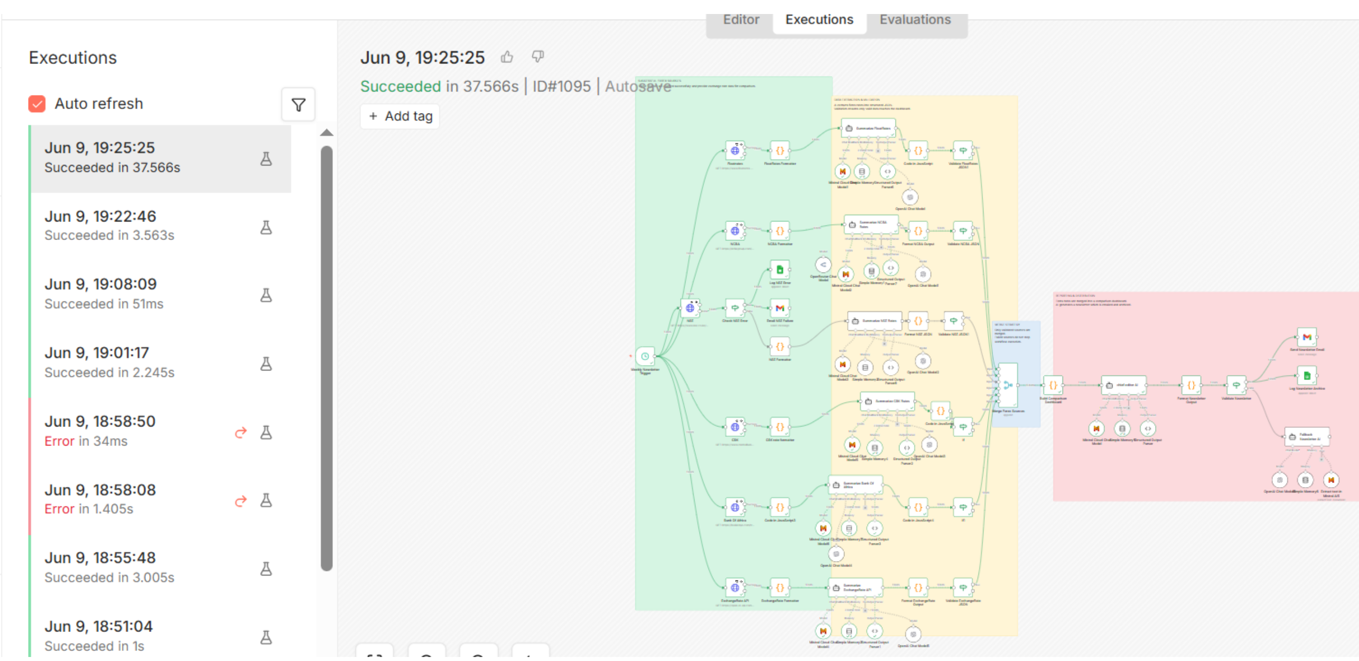
- Detect workflow-level failures
- Log operational errors
- Send automated notifications
- Improve operational visibility

The Error Handling Workflow performs:

1. Error detection using Error Trigger node
2. Logging into Google Sheets
3. Gmail error notification delivery



6. Screenshot of Execution Logs



7. Step-by-Step Workflow Execution

1. Scheduled trigger activates workflow
2. Financial sources are queried independently
3. Source formatter nodes clean and normalize outputs
4. Mistral AI summarizes financial intelligence
5. Structured output parsers validate AI outputs
6. Validation nodes check JSON reliability
7. Valid outputs proceed to merge layer
8. Financial comparison dashboard is generated
9. Chief Editor AI generates final newsletter
10. Newsletter validation is performed
11. Newsletter is emailed through Gmail
12. Newsletter is archived into Google Sheets
13. Fallback AI handling activates where necessary
14. Error Handling Workflow activates during failures

8. Integrations, APIs, and Tools Used

Workflow Automation

- n8n

AI Processing

- Mistral AI

Email Distribution

- Gmail

Data Logging & Archival

- Google Sheets

Financial Data Sources

- FloatRates
- NCBA
- NSE
- CBK
- Bank of Africa
- ExchangeRate API

Programming & Data Processing

- JavaScript Code Nodes
- Structured Output Parser Nodes

9. Error Handling & Recovery Mechanisms

Continue-On-Fail Strategy

HTTP Request nodes were configured to continue execution even if individual financial sources fail.

Purpose:

- Prevent total workflow collapse
- Support graceful degradation

Branch-Level Isolation

Each financial source operates independently.

Purpose:

- Isolate failures
- Prevent cascading workflow errors

Structured Output Validation

AI outputs are validated before downstream processing.

Purpose:

- Prevent malformed AI outputs
- Improve machine reliability

Merge Validation Strategy

Only validated outputs proceed into the merged workflow path.

Purpose:

- Prevent invalid data propagation
- Improve reporting accuracy

Error Trigger Workflow

A dedicated Error Handling Workflow captures workflow-level failures.

Purpose:

- Improve operational monitoring
- Automate error logging
- Automate error notifications

Operational Logging

Workflow execution outputs and newsletter archives are stored in Google Sheets.

Purpose:

- Governance visibility
- Historical tracking
- Operational auditing

AI Fallback Model Strategy

Fallback AI handling was implemented within the workflow to improve operational resilience during AI processing failures.

Purpose:

- Prevent complete workflow interruption during AI-related failures
- Maintain workflow continuity
- Improve production reliability
- Support graceful degradation during model issues

The fallback strategy activates when:

- AI responses fail validation

- Structured outputs are malformed
- AI summarization fails
- Newsletter generation encounters processing issues

Fallback AI nodes were configured within the reporting and newsletter generation layer to ensure the workflow can continue generating outputs even when the primary AI processing path encounters issues.

This implementation improves:

- Operational reliability
- AI fault tolerance
- Workflow continuity
- Production-grade resilience